Bérince S. R. Hounsouvo

Ph. D Candidate in Forest sciences, Specializing in Tree-Fungi-Soil Interactions | Université Laval
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Education

Ph.D. in Forest Sciences 2023 - 2026
Department of wood and forest sciences, Université Laval, Québec Canada
Thesis: Selection of boreal woody species with lateral root architecture for technosol-based revegetation of acid mine drainage sites
Supervisor: Prof. Damase P. Khasa
Co supervisor: Prof. Antoine Karam

Master's degree, Forest Sciences and Techniques 2018 - 2020
Faculty of Agronomy, University of Parakou, (Benin)
Thesis: Effect of ectomycorrhizal inoculation and organic and mineral amendments on juvenile growth of nursery plants of Afzelia africana, Pterocarpus erinaceus & Khaya senegalensis.
Supervisor: Prof. Christine Ouinsavi

Bachelor's degree, Natural Resource Management 2014 - 2017
Faculty of Agronomy, University of Parakou, (Benin)
Thesis: Optimal sampling technique for plot-based biodiversity measurement in the Sudanian zone: the case of square and circular plots
Supervisor: Prof. Nourou S. Yorou

Additional Training

[6] Hierarchical models and Bayesian inference for the natural sciences, Winter 2026, Université Laval, QC, Canada
[5] Big data processing and functional predictions in Biology - SymbioSol workshop 2025, Quebec, Canada.
[4] NSERC CREATE Program on Nature-Based Solutions for Ecosystem Restoration - NASER Summer School 2025 [in person] . Université Laval, Quebec, Canada. From April 28 to May 09, 2025

[3] Programming with R for data analysis [in person]. Université Laval, Québec, Canada.

[2] Introduction to Unix Shell [online, UNX101]. Calcul Québec, Quebéc, Canada. February 2, 2024

[1] Data analysis in bioinformatics: exploring computational biology [in person]. Université Laval, Quebec, Canada. From April 28 to May 09, 2025

Research Interests

- Forest biology
- Mycorrhizal symbioses
- Forest ecophysiology
- Restoration ecology
- Urban forestry

Academic and Research Experiences

Research Assistant, Forest Biology 2024 - present
Department of wood and forest sciences, Université Laval, Québec Canada.
- Assist a PhD candidate in designing a forest field experiment
- Participate in ECM (ectomycorrhizal) seedling inoculation
- Manage fungal strain collection

Doctoral Researcher, Forest biology 2023 - present

Department of wood and forest sciences, Université Laval, Québec Canada.

- Develop literature reviews on our scientific knowledge of tree-soil and tree-fungi interactions.
- Design and carry out advanced experimental protocols in forestry and restoration practices.
- Build and run advanced statistical models using R and JAGS.

Research Associate, Forest Biology 2025 - present

Forest Studies and Research Laboratory, University of Parakou, Parakou (Benin)

- Design and conduct forest experimentation projects
- Conduct analyses using linear mixed-effects models in R
- Write scientific and field mission reports

Teaching Assistant, Forest Biology Winter 2025

Department of soils and agro-food engineering, Université Laval, Québec Canada

- Co-supervise a master's student in the design and implementation of a forest experiment
- Provide scientific guidance on experimental design and field data collection

Research Assistant, Tropical Mycology 2017 - 2018

Research Unit in Tropical Mycology and Plant-Fungi-Soil Interactions, Parakou (Benin)

- Conduct surveys of macro-fungi in open forests and savannas
- Evaluate fungal diversity indices (α and β diversity) using R software
- Manage fungi specimens in the Mycological Herbarium collection at the University of Parakou

Publications

[5] **Hounsouvo BSR** et al., Using mycorrhizal inocula for tree in Africa: an integrative conceptual framework and research roadmap (in preparation for Biological reviews).

[4] **Hounsouvo BSR**, Khasa DP, Karam A and Lamhamedi MS. Tree root development and architecture at post-mining sites: A comprehensive systematic review (in preparation for STOTEN).

[3] **Hounsouvo BSR**, Orou Sourou Z, Ayena JIK, and Dramani R. Socio-Agronomic Determinants of Farmland Access in Vegetable Production Systems in Northern Benin: Evidence from Bassila. Agriculture and Human Values (submitted). https://doi.org/10.21203/rs.3.rs-6614725/v1

[2] **Hounsouvo BSR**, Houehanou TD, Gouwakinnou GN, Hadonou CJ, and Ouinsavi CAIN. (2022). Comparative Effect of Ectomycorrhizal Inoculation, Mineral and Organic Amendments and their Interactions on Juvenile Growth of Afzelia africana in the Nursery. Journal of Tropical Plant Physiology, 14(1), 1-12. 022):1-12 https://doi.org/10.56999/jtpp.2022.14.1.18

[1] Habakaramo MP, Tchan KI, Hegbe ADMT, Abohoumbo G, **Hounsouvo BSR**, Tchemagnon O, Dramani R, and Yorou NS (2022). Sampling techniques for the optimal measurements of macromycetes diversity in the Soudano-Guinean ecozone (West Africa). International Journal of Biodiversity and Conservation, 14(3), 128-138. https://doi.org/10.5897/IJBC2020.1404

Scientific Communications

[4] Root System Strategies of Picea mariana and Larix laricina for Revegetation of Acid Mine Drainage Sites in Quebec. Journée étudiante de Centre SÈVE, Quebec, Canada. [Poster]

[3] Variation génétique de la croissance racinaire chez les plants d'épinette noire et de mélèze laricin. Colloque forestier 2025. L'office des producteurs de plants forestiers du Québec (OPPFQ)Quebec, Canada. [Oral presentation]

[2] Genetic Control and Repeatability of Root Growth in Seedlings of EPN and MEL, Two Species with Contrasting Strategies Along the Root Economics Spectrum. Interlab 2025. Institute of Integrative Biology and Systems, Université Laval, Quebec, Canada. [Oral presentation]
[1] Root System Strategies of Picea mariana and Larix laricina for Revegetation of Acid Mine Drainage Sites in Quebec. Journée étudiante de l'IBIS (15e ed), Institute of Integrative Biology and Systems, Université Laval, Quebec, Canada. [Poster]

Awards - Grants - Fellowships ($84,650)

Doctoral degree - $81,400

[9] J. André Fortin grant in Forest Mycology and Forest Soil Biology: Influence of root morphology on the mycorrhizal community of boreal woody species (2026) - $2500.
[8] Doctoral thesis writing grant, Faculty of Graduate and Postdoctoral Studies, Université Laval (2026) - $5000.
[7] Best PhD progress award (37RAPF), Faculty of Graduate and Postdoctoral Studies, Université Laval (2025) - $3700.
[6] Participation Grant for the Student Day 2025, Centre SÈVE: Root System Strategies of Picea mariana and Larix laricina for Revegetation of Acid Mine Drainage Sites in Quebec (2025) - $1000
[5] Doctoral project completion Award (37EXAD), Faculty of Graduate and Postdoctoral Studies, Université Laval (2020) - $6200

[4] Award for doctoral project preparation, Faculty of Graduate and Postdoctoral Studies, Université Laval (2024) - $3000
[3] Doctoral fellowship, Department of wood and forest sciences, Université Laval (2023-2026) - $60,000

Master's degree - $750

[2] Master's fellowship, University of Parakou, Benin Republic (2019-2020) - $750

Bachelor's degree - $2500

[1] Bachelor's scholarship, Minister of Higher Education and scientific research, Benin Republic (2014-2017) - $2500

Professional membership

International Society of Root Research since 2025
Society of Ecological restoration since 2024
British Ecological Society since 2019

Mentoring

Graduate co-advisor

[3] Franel Affolabi 2025. Sélection des espèces à racines traçantes dans le contexte agroforestier post minier. Master's degree, Université Laval, Québec, Canada.
[2] Justine Daviault 2025. Pratiques de revégétalisation des sites miniers générateurs de drainage minier acide au québec : acquis, opportunités et défis. Master's degree, Université Laval, Québec, Canada.

Undergraduate co-advisor

[1] Osé Zoundoh 2020. Effets de différentes doses d'amendements organique et minérale sur la croissance juvénile de Afzelia, khaya et Pterocarpus. Bachelor's degree, Universitý of Parakou, Parakou, Benin